

UNSUPERVISED LEARNING METHOD

ASSIGNMENT: ASSOCIATION RULE LEARNING

Association Rule Learning Using the Groceries Dataset

Dataset: Groceries Dataset
Method: Association Rule Learning
Algorithm: Apriori
Software: Python
Transactions: 9,835
Different Products: 169

Student Name: _____

Course/Module: _____

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Introduction

Association Rule Learning is an unsupervised learning technique used to discover relationships between items that frequently occur together in transactions.

In this assignment, the Groceries dataset was analyzed to identify frequent itemsets and association rules that describe customer purchasing behavior.

The analysis was performed using Python. During data preparation, completely empty rows and columns were removed, product names were standardized by removing unnecessary spaces and converting them to lowercase, and duplicate products within individual transactions were removed.

The cleaned transactions were then converted into a one-hot encoded transaction matrix. The Apriori algorithm was used to identify frequent itemsets with a minimum support of 1%. Association rules were subsequently generated using a minimum confidence threshold of 10%.

Question 1: Understand the Data

1.1 Number of Transactions

The cleaned dataset contains:

9,835 transactions

Each row represents one customer transaction.

1.2 Number of Different Grocery Items

The dataset contains:

169 different grocery items

1.3 Ten Most Frequently Purchased Items

The ten most frequently purchased grocery items are shown in Table 1.

Table 1: Top 10 Most Frequently Purchased Items

Rank	Product	Frequency
1	Whole milk	2,513
2	Other vegetables	1,903
3	Rolls/buns	1,809
4	Soda	1,715
5	Yogurt	1,372
6	Bottled water	1,087
7	Root vegetables	1,072
8	Tropical fruit	1,032
9	Shopping bags	969
10	Sausage	924

The results show that whole milk was the most frequently purchased product, appearing in 2,513 transactions.

Question 2: Frequent Itemsets

The transactions were transformed into a one-hot encoded matrix with 9,835 transactions and 169 product columns.

The Apriori algorithm was applied with a minimum support threshold of 0.01 (1%). Only itemsets containing at least two products were considered for the requested frequent-itemset analysis.

2.1 Top 10 Frequent Itemsets

Table 2: Top 10 Frequent Itemsets

Itemset	Support
Other vegetables, whole milk	0.074835
Rolls/buns, whole milk	0.056634
Whole milk, yogurt	0.056024
Root vegetables, whole milk	0.048907
Other vegetables, root vegetables	0.047382
Other vegetables, yogurt	0.043416
Other vegetables, rolls/buns	0.042603
Tropical fruit, whole milk	0.042298
Soda, whole milk	0.040061
Rolls/buns, soda	0.038332

2.2 Itemset with the Highest Support

The itemset with the highest support is:

{other vegetables, whole milk}

with:

$$Support = 0.0748 = 7.48\%$$

Support is calculated as:

$$Support(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

Therefore, 7.48% of all transactions contain both other vegetables and whole milk.

2.3 Meaning of Support

Support indicates how frequently an itemset occurs in the complete transaction dataset. A higher support value means that the itemset appears in a larger proportion of transactions.

In this analysis, other vegetables and whole milk form the most frequently occurring two-product itemset among the identified itemsets.

Question 3: Association Rules

Association rules were generated from the frequent itemsets using a minimum confidence threshold of 10%.

The ten rules with the highest confidence are presented in Table 3.

Table 3: Top 10 Association Rules

Antecedent	Consequent	Support	Confidence	Lift
Citrus fruit, root vegetables	Other vegetables	0.010371	0.586207	3.029608
Root vegetables, tropical fruit	Other vegetables	0.012303	0.584541	3.020999
Curd, yogurt	Whole milk	0.010066	0.582353	2.279125
Butter, other vegetables	Whole milk	0.011490	0.573604	2.244885
Root vegetables, tropical fruit	Whole milk	0.011998	0.570048	2.230969
Root vegetables, yogurt	Whole milk	0.014540	0.562992	2.203354
Domestic eggs, other vegetables	Whole milk	0.012303	0.552511	2.162336

Antecedent	Consequent	Support	Confidence	Lift
Whipped/sour cream, yogurt	Whole milk	0.010880	0.524510	2.052747
Rolls/buns, root vegetables	Whole milk	0.012710	0.523013	2.046888
Other vegetables, pip fruit	Whole milk	0.013523	0.517510	2.025351

For an association rule:

$$A \rightarrow B$$

support measures how frequently the complete combination occurs in the dataset, confidence measures how frequently B occurs when A occurs, and lift measures the strength of the relationship relative to the baseline frequency of the consequent.

Question 4: Understanding the Measures

Three association rules were selected for detailed interpretation.

4.1 Rule 1

$$\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$$

- **Support = 0.0104 (1.04%):** About 1.04% of all transactions contain citrus fruit, root vegetables, and other vegetables together.
- **Confidence = 0.5862 (58.62%):** Among transactions containing citrus fruit and root vegetables, 58.62% also contain other vegetables.
- **Lift = 3.0296:** Since the lift is greater than 1, the rule indicates a positive association between the antecedent and the consequent.

4.2 Rule 2

$$\{\text{root vegetables, tropical fruit}\} \rightarrow \{\text{other vegetables}\}$$

- **Support = 0.0123 (1.23%):** About 1.23% of all transactions contain the three products together.
- **Confidence = 0.5845 (58.45%):** Among transactions containing root vegetables and tropical fruit, 58.45% also contain other vegetables.
- **Lift = 3.0210:** The lift is greater than 1, indicating a positive association.

4.3 Rule 3

$$\{\text{curd, yogurt}\} \rightarrow \{\text{whole milk}\}$$

- **Support = 0.0101 (1.01%)**: About 1.01% of all transactions contain curd, yogurt, and whole milk together.
- **Confidence = 0.5824 (58.24%)**: Among transactions containing curd and yogurt, 58.24% also contain whole milk.
- **Lift = 2.2791**: Since the lift is greater than 1, the rule represents a positive association.

4.4 Summary of Measures

Table 4: Meaning of Association Rule Measures

Measure	Meaning
Support	Proportion of all transactions containing the complete itemset.
Confidence	Proportion of transactions containing the antecedent that also contain the consequent.
Lift	Strength of the association relative to the baseline frequency of the consequent.

Question 5: Interesting Rules

The generated rules were compared according to their confidence, lift, and support.

5.1 Rule with the Highest Confidence

$$\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$$

- Support = 0.0104 (1.04%)
- Confidence = 0.5862 (58.62%)
- Lift = 3.0296

This is the rule with the highest confidence in the generated association-rule results.

5.2 Rule with the Highest Lift

$$\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$$

- Support = 0.0101 (1.01%)
- Confidence = 0.1889 (18.89%)
- Lift = 3.3723

This rule has the highest lift in the generated results.

5.3 Rule with the Highest Support

$$\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$$

- Support = 0.0748 (7.48%)
- Confidence = 0.3868 (38.68%)
- Lift = 1.5136

This rule has the highest support among the generated association rules.

5.4 Are They the Same Rule?

No. The three measures identify different rules.

The highest-confidence rule is different from the highest-lift rule, and both are different from the highest-support rule. This shows that support, confidence, and lift measure different characteristics of an association rule.

A rule can occur frequently without having the highest confidence or lift. Similarly, a rule can have high lift while occurring in a relatively small proportion of all transactions.

Question 6: Interpret a Rule

The selected rule is the rule with the highest lift:

$$\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$$

In simple language, this rule means that customers who purchase curd also tend to purchase whole milk and yogurt.

The support is:

$$0.0101 = 1.01\%$$

This means that 1.01% of all transactions contain curd, whole milk, and yogurt together. The confidence is:

$$0.1889 = 18.89\%$$

This means that 18.89% of transactions containing curd also contain whole milk and yogurt.

The lift is:

$$3.3723$$

Since the lift is greater than 1, the rule indicates a positive association. The value indicates that the consequent occurs substantially more often with the antecedent than would be expected from its baseline occurrence.

The association can therefore be described as relatively strong in terms of lift. However, the support is only 1.01%, meaning that the complete combination occurs in a relatively small proportion of all transactions.

Question 7: Business Application

Association rules can be used by grocery stores for product placement, bundling, and recommendation systems.

7.1 Two Products to Place Close Together

The two products that could be placed close together are:

Other vegetables and whole milk

This is based on the rule:

$$\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$$

The rule has:

- Support = 0.0748 (7.48%)
- Confidence = 0.3868 (38.68%)
- Lift = 1.5136

The relatively high support means that this combination occurs frequently in the transaction dataset.

7.2 One Possible Product Bundle

A possible bundle is:

Curd + Whole milk + Yogurt

This is based on the rule:

$$\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$$

which has the highest lift in the generated rules:

$$Lift = 3.3723$$

7.3 Example of Product Recommendation

If a customer purchases curd, the store could recommend whole milk and yogurt.

For example:

“Customers who buy curd may also be interested in whole milk and yogurt.”

This recommendation is based on the observed association in the transaction data.

Question 8: Understanding Association

8.1 Does Association Prove Causation?

No. Association does not prove causation.

Association rules show that products tend to occur together in transactions. However, they do not prove that purchasing one product causes the purchase of another product.

8.2 Why Can a Rule Have High Confidence but Low Lift?

A rule can have high confidence but low lift when the consequent is already very common in the dataset.

For example, if a product is purchased frequently by many customers, then many transactions containing another product may also contain that common product. This can result in high confidence even when the additional association is not particularly strong.

Therefore, confidence should be considered together with lift.

8.3 What Does Lift Greater Than 1 Indicate?

A lift greater than 1 indicates a positive association.

It means that the consequent occurs more frequently when the antecedent is present than would be expected from the overall frequency of the consequent.

For example:

$$\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$$

has:

$$Lift = 3.3723 > 1$$

Therefore, this rule represents a positive association.

Conclusion

This assignment applied Association Rule Learning to the Groceries dataset using the Apriori algorithm.

After cleaning the data, the dataset contained 9,835 transactions and 169 different grocery products. Whole milk was the most frequently purchased product, appearing in 2,513 transactions.

The frequent-itemset analysis showed that the combination of other vegetables and whole milk had the highest support among the identified itemsets, with a support of 7.48%.

Several association rules were generated. The rule with the highest confidence was:

$$\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$$

with confidence of 58.62%.

The rule with the highest lift was:

$$\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$$

with lift of 3.3723.

The rule with the highest support was:

$$\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$$

with support of 7.48%.

The analysis demonstrates that support, confidence, and lift provide different but complementary information about purchasing behavior. These measures can be applied to product placement, product bundling, and product recommendation. However, association should not be interpreted as proof of causation.

Question A: Python Code and Submission Materials

The complete Python implementation is submitted separately as the Jupyter Notebook `HW_1.ipynb`.

The notebook contains the following main stages:

1. Loading the Groceries dataset.
2. Removing completely empty rows and columns.
3. Cleaning and standardizing product names.
4. Removing duplicate products within transactions.
5. Saving the cleaned dataset.
6. Calculating the number of transactions and unique products.
7. Identifying the ten most frequently purchased products.
8. Transforming transactions into a one-hot encoded matrix.
9. Generating frequent itemsets using the Apriori algorithm.
10. Generating association rules using confidence.
11. Comparing rules using support, confidence, and lift.
12. Interpreting the resulting rules and their business applications.

The final submission therefore consists of this short report together with the Jupyter Notebook containing the executable Python code.